The Nature of ROC Practices in Eyewitness Memory Research

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This is a preprint and has not yet undergone peer review.

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Abstract

Eyewitness memory research has reformed police practices and policy and is sometimes relied upon in legal proceedings. Due to the practical implications derived from this research, it is imperative to evaluate how practical recommendations are postulated. To assess the practical relevance of research, effect sizes and its interpretation play a pivotal role. In this study, we examined how the frequently used effect size Area Under the Curve (AUC) obtained via Receiver Operating Characteristic (ROC) curves are used and interpreted in eyewitness memory research. Via a systematic literature search, we identified 157 eyewitness memory related articles that conducted ROC curve analyses resulting in 1580 AUC values. Approximately 90% of the studies relied on statistical significance to interpret the AUC values. This practice can be problematic as the sample sizes in many studies were rather large which can lead to statistically significant effects that are trivial. Also, more than half of the studies did not report 95% CIs for their AUC values. Finally, power analyses were frequently not conducted and if they were they were, they were oftentimes not reproducible. To improve the practical inferences of eyewitness memory research, we highlight the need of establishing a smallest effect size of interest, focusing on 95% CIs, and conducting reproducible power analyses.

The Nature of ROC Practices in Eyewitness Memory Research

Don't dichotomize variables. Unless you do. - *Patrick Curran & Gregory Hancock, Quantitude (2023)*

In the National Registry of Exonerations (2023), legal cases can be reviewed revealing possible disastrous consequences of mistaken eyewitness memory.[1] For example, mistaken eyewitness identifications played a pivotal role in 331 (56%) out of the 592 DNA exonerations in the United States. These exonerations have fueled the attention of experimental psychologists to conduct lab studies to examine system variables (e.g., type of lineup) or estimator variables (e.g., intoxication) that may improve or impede eyewitness memory. Because of the practical nature of eyewitness memory studies, findings from this field have led to reforms in police practices and policy such as double-blind testing and collecting confidence statements (National Research Council, 2009; Wells et al., 2020). Furthermore, psychologists are sometimes consulted as expert witnesses in court to evaluate eyewitness memory and identifications procedures (Pezdek, 2020).

However, is has been argued that psychological research can only enter the courtroom, if the research shows signs of replicability, generalizability, and practically relevance (Otgaar et al., 2022). Although all three constructs are vital to obtain confidence in psychological research (e.g., Chin & Neal, 2023; Otgaar et al., 2023), for this study, we exclusively focused on the importance of practical relevance. Specifically, we examined the practical relevance of eyewitness memory research by critically evaluating often-used effect sizes (i.e., magnitude of a phenomenon; Cohen, 1988) in this area of research. Interestingly, the practical relevance of eyewitness memory research by focusing on effect sizes is still in its infancy (Chin et al.,

[1] [1] In the current study, when we refer to eyewitness memory research we also mean eyewitness identification research.

2023) and only recent studies have urged the need for establishing which effect sizes are practically relevant (e.g., Otgaar et al., 2023; Riesthuis et al., 2021).

A frequently used effect size in eyewitness memory studies is the area under the curve (AUC; Smithson, 2023) which is obtained via Receiver Operating Characteristic (ROC) curve analyses. Interestingly, in other fields (i.e., health and medical fields), questionable research practices have been detected when ROC curve analyses and AUCs were reported (White et al., 2023). For example, White and colleagues (2023) found an overinflation of certain AUC values which might have been caused by researchers re-analyzing their data until they reach a desired result. Because of strong reliance on ROC curves and AUCs in eyewitness identification research (e.g., Lampinen, 2016; Wixted & Mickes, 2018), the question can be raised on how ROC curves and their corresponding AUCs in eyewitness memory research are currently applied and interpreted. Hence, the crux of the present study was to examine the application of ROC curve analyses and AUC values in eyewitness memory research, with a specific focus on its effect size interpretation.

ROC Curve Analyses in Eyewitness Memory Research

ROC curve analyses can provide crucial insights in terms of empirical discriminability[2] (Smith & Neal, 2021; Wixted & Mickes, 2018). Empirical discriminability refers to the extent an eyewitness can accurately differentiate between guilty (or old/studied items) and innocent suspects in a lineup (or new/non-studied items; Gronlund et al., 2012). Simply put, the higher the discriminability of a procedure (e.g., type of lineup) the better. The concept underlying ROC curve analyses in eyewitness memory studies is rather straightforward (for an extended explanation on ROC curve analyses see Brady et al., 2023).

For instance, imagine a typical eyewitness memory study wherein participants watch a mock crime and later on their memory is tested through a confidence-based recognition

[2] We only focus on empirical discriminability due to the focus on practical relevance and not theory.

memory task (e.g., lineup; “on a scale from 1-6, how confident are you that that the identified individual is the perpetrator?”; *1 = not confident at all, 6 = very confident*). Based on these data, hit rates (i.e., correct identification rate) and false positive rates (i.e., mistaken identification rate) can be calculated at each confidence interval or threshold. In other words, if participants used a low threshold of “*1 = not confident at all*” (i.e., indicating identified with this confidence or higher) for indicating that they recognize the perpetrator or innocent suspect, hit and false positive rates are calculated. Then, the hit and false positive rates for a slightly more conservative threshold (i.e., *2 = somewhat confident*) can be calculated and this is done for each threshold. This results in a ROC curve wherein for each level of confidence the hit and false alarm rates are plotted on y-axis and x-axis, respectively, with confidence levels being depicted from conservative (left side of the graph) to liberal (right side of the graph; see Figure 1; Mickes et al., 2012).

Then, the area under the ROC curve can be estimated (i.e., AUC or AUROC). The AUC is a value between 0 and 1 (or 0 to 100%) and indicates how well the model performed. Perfect discriminability is characterized with a ROC curve passing through the upper left corner which then has an AUC value of 1 (or 100%; see black line Figure 1). If the ROC curve is diagonal, it means that the model performs at chance level which then has an AUC of .5 (see blue line figure 1). However, typically a ROC curve/AUC value will fall somewhere between these two extremes which brings the burden upon the researcher to make meaningful interpretations about the practical relevance of this value (i.e., see red line in Figure 1). One way to interpret the AUC is as the probability that a randomly chosen datapoint of a target present lineup distribution has higher confidence associated with it than a randomly chosen datapoint of a target absent lineup distribution (Hanley & McNeil, 1982). For instance, in the case of the typical AUC (i.e., red line figure 1), there is a 78% chance that a random chosen

datapoint of a target present lineup distribution has higher confidence associated with it than a randomly chosen datapoint of a target absent lineup distribution.

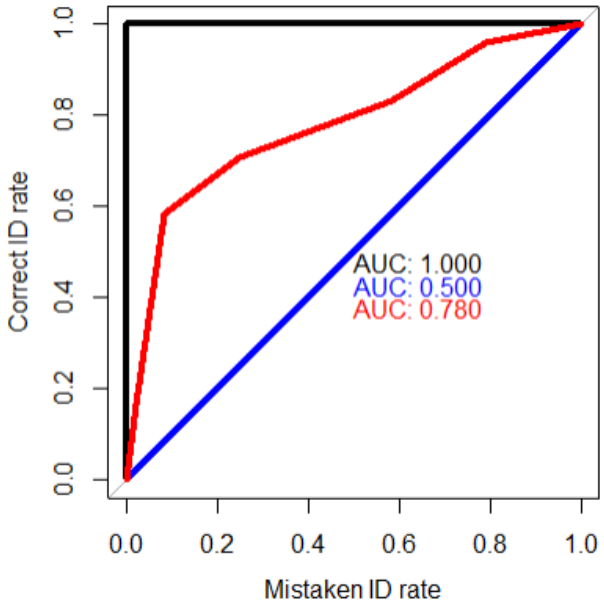


Figure 1. Examples of ROC curves. Black is perfect discriminability, blue is chance level discriminability, and red is a more typical ROC curve.

Another frequently used statistic in eyewitness identification research, is the partial AUC (pAUC). The pAUC captures only a part of the ROC curve because it takes into account that the false positive rate is limited by, for example, the number of persons in a lineup. For instance, if a fair six-person lineup is used, the maximum false positive rate is not 1 but is .167 (1/6; Wixted & Mickes, 2018). The pAUC can be calculated for this region and can then be compared to other pAUCs to examine which model yields higher discriminability (see figure 2). However, the question is how such pAUC and AUC values should be meaningfully interpreted in terms of its practical relevance.

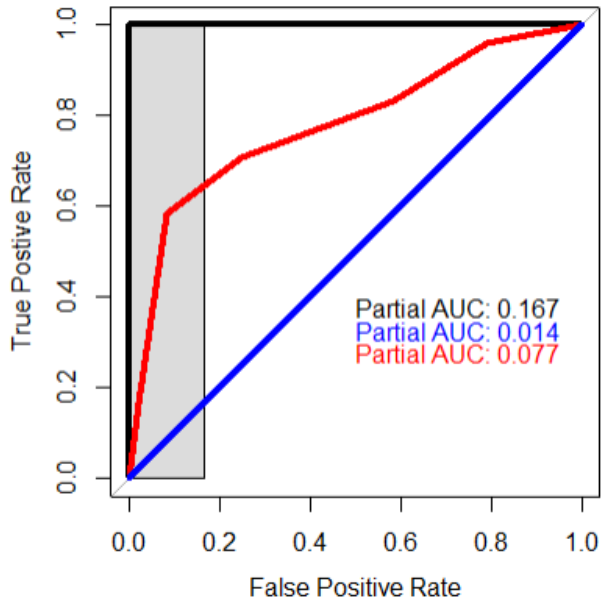


Figure 2. Example of pAUC values. Maximum pAUC value depends on the maximum false positive rate which in this example is set to .167.

ROC curve and the Smallest Effect Size of Interest

Besides perfect and chance-level discriminability, how should ROC curves and (p-)AUC values be meaningfully interpreted? In other words, how do we know when a certain type of lineup is superior to other lineups and when can practical recommendations be made based on certain ROC curves or AUC values? One possibility is to visually compare the ROC curves and examine whether one is entirely above another. If this happens, it could be concluded that a certain lineup superior to another one. Although visual inspection and comparison of ROC curves is important, it should be established whether these differences are not too small and possibly due to random variation. It is also possible to statistically compare ROC and (p-)AUC values which can be done in for example the R package "pROC" (Robin et al., 2011).

However, relying solely on statistical significance is problematic for several reasons (Gardner & Altman, 1986). First, relying on statistical significance without visually examining the ROC curves may oversimplify the conclusion that one procedure is better than the other (Obuchowski, 2005). For instance, ROC curves might overlap which complicates

the interpretation even when they are statistically significantly different (see figure 3; Moise et al., 1988). Second, with small sample sizes, the obtained estimates can be inflated, unreliable and difficult to replicate (Button et al., 2013). However, this may not pose a serious threat in eyewitness identification studies because researchers have noted that large sample sizes are needed to obtain precise estimates in ROC curve analyses (Kovera & Evelo, 2021), even though this depends on the context and parameters of interest for each study (Obuchowski, 2005). Possibly more problematic is that with increasing sample sizes, statistical significance can be found for trivial effects (Anvari & Lakens, 2021). For example, with large enough sample sizes a statistically significant ($p < .05$) difference of .01 may be detected between two (p-)AUC values but what this means in practical terms is difficult to grasp. In other words, researchers might conflate statistical significance with practical relevance which is a common misconception among memory researchers (Riesthuis et al., 2021).

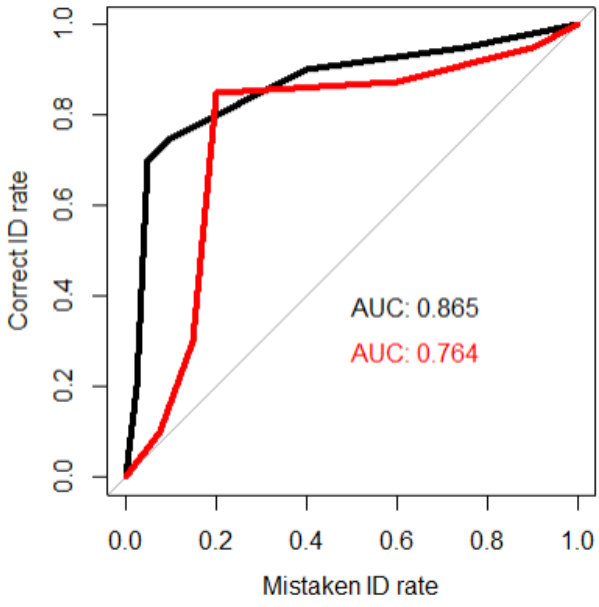


Figure 3. Example of overlapping ROC curve analyses wherein differences in AUC are statistically significant ($z = 15.1$, $p < .001$ (using "delong" method: $D = 14.9$, $p < .001$)

Preferably, ROCs and AUC values are regarded as a type of effect size indicating the magnitude of an effect which is in turn interpretated in the context of the study at hand and

the associated costs and benefits (Baker et al., 2014; Cohen, 1988; Smithson, 2023). Similar to Cohen's benchmarks (e.g., Cohen's *d* of .2, .5, and .8 being small, medium, and large effects, respectively; 1988), researchers have proposed benchmarks to interpret the magnitude of AUC values wherein .7 - .8 are considered fair/acceptable, .8 - .9 are considered good, and above .9 are considered excellent (Mandreker, 2010). However, just as with Cohen's benchmarks, no justification exists for these subjective benchmarks (Panzarella et al., 2021). Instead, it is preferred that values should be evaluated based on the context of the study and the costs and benefits associated with it (Baker et al., 2014; Lakens et al., 2018; Riesthuis et al., 2022; White et al., 2023). Due to strong reliance on statistical significance (e.g., Riesthuis et al., 2021) and benchmarks (e.g., Correl et al., 2020) to interpret the magnitude of effects in psychological research, we examined how ROC curves and AUCs are used and interpreted in eyewitness memory research.

The Present Study

In the present study, we examined how eyewitness memory researchers use and interpret ROC curve analyses and AUCs. Through a literature search, we identified 157 articles that implemented ROC curve analyses in eyewitness memory research. Then, we extracted their AUC values, 95% CIs (if reported), and how they were interpreted. Research examining clinical prediction models found an excess of AUC values above and shortfalls right before the proposed AUC benchmarks of .7 - .8 (i.e., acceptable), .8 (i.e., good), and .9 (excellent) for (White et al., 2023). We investigated whether similar excesses and shortfalls were observed in the eyewitness memory literature. Moreover, we examined the interpretation of the AUC values in terms of taking its variation into account (i.e., reporting 95% CIs) and its magnitude. Last, we looked at whether and how power analyses were conducted to obtain insights into how eyewitness memory research considers which effects are meaningful at the designing stage of a project.

Method

Literature Search

The Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA; Page et al., 2021) workflow was used to present the steps of the literature search (Figure 4). We preregistered our study on the Open Science Framework (OSF; https://osf.io/fxu28). We examined the databases PsycINFO, PsycArticles, and Google Scholar for our literature search. For our literature search, we used the following keywords to appear anywhere in a full article: ("eyewitness memory" OR "eyewitness identification" OR “eyewitness ID”) AND (“area under the receiver operating characteristic curve” OR “area under the curve” OR "Receiver Operating Characteristic" OR “AUC” OR “AUROC” OR "ROC"). We included all articles that were published before 4th of January 2024 which is the date that we conducted the literature search in the databases.

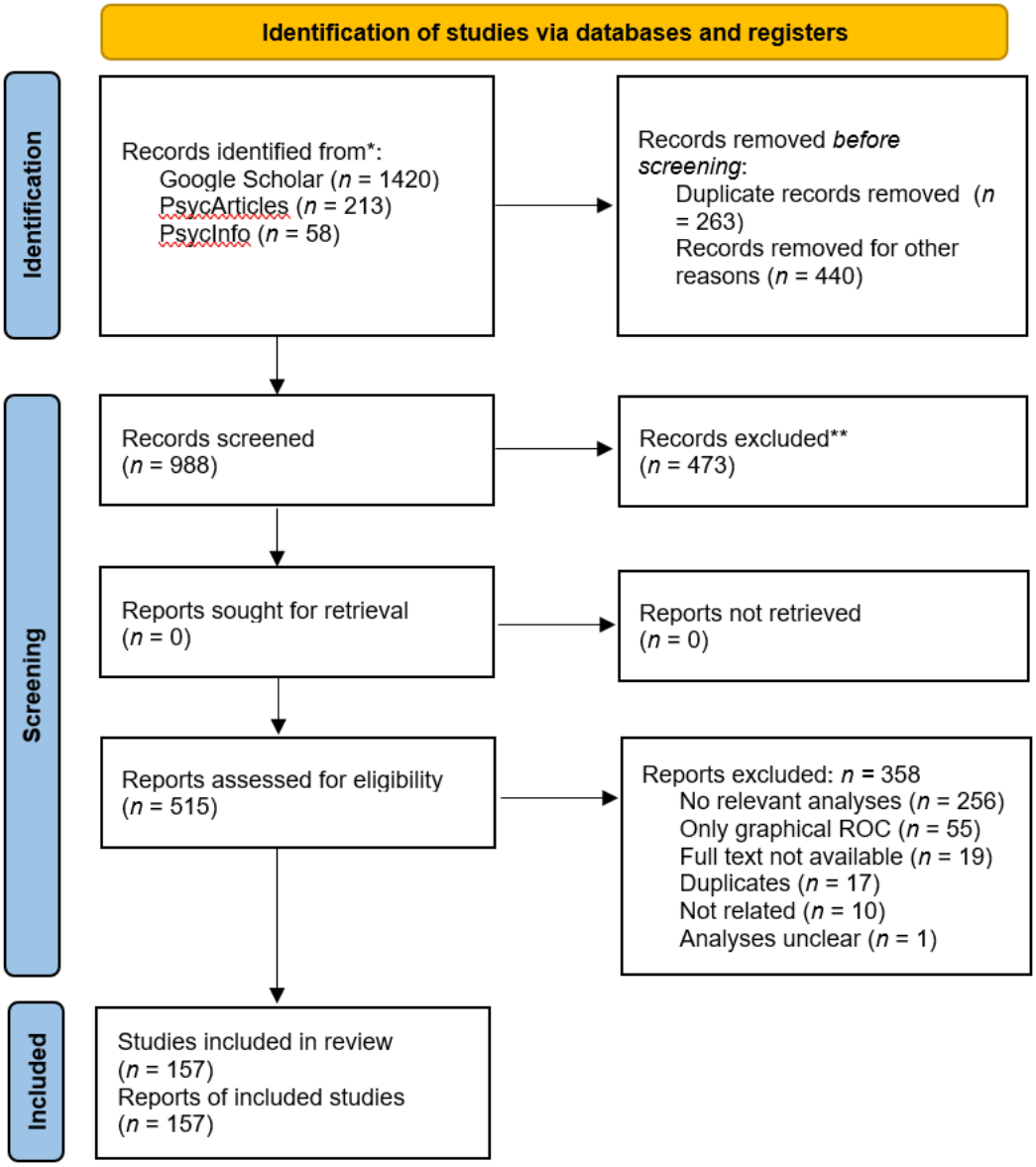


Figure 4. PRISMA workflow chart of literature search.

Our first search led to 1691 articles ($n_{\text{google Scholar}}$[3] = 1420, $n_{psycArticles}$ = 213, $n_{psycInfo}$ = 59). In the first round, we included all articles that were either an article, preprint, or thesis. Meta-analyses, reviews, commentaries, blogs, chapters, books, and presentations were excluded (see OSF https://osf.io/dgxsz/ for literature search for each database). We only included studies that were related to eyewitness identification or memory research. We

[3] We were only able to go through 980 articles of Google Scholar because Google Scholar does not allow its users to go through results beyond that. We sorted them by relevance and included the first 980 articles in our literature search.

excluded all duplicates across the databases and articles that did not report an experiment (e.g., review) or re-analyses of previous data. To include as many articles as possible in the second round, both authors independently indicated which articles would advance to the next round. Articles that were marked as "include" by one of the raters would proceed to the second round even if the other author indicated that it should not. After excluding articles based on these criteria, 515 articles remained. In the second round, the first author conducted a full-screening round in which articles were excluded if they were not related to eyewitness identification or memory research, full text was not available, did not conduct relevant analyses or did not report them (i.e., ROC curve analyses, AUC), or when they only provided the graphical information (i.e., ROC curve plot with no further information; see Figure 4). This resulted in a final set of 157 articles that were coded and included in the data analysis.

Coding

One coder (first author) independently coded all 157 studies and extracted which type of AUC value was reported (i.e., partial, full, difference), (p-)AUC values, its 95% CIs (if reported), interpretation of the (p-)AUC values, p-values, sample size, whether a power analysis was conducted and if it was reproducible. All included articles and the extracted statistics are available on OSF (https://osf.io/dgxsz/). Several checks were implemented such as double-checking off seeming (p-)AUCs (e.g., relatively large or small) and checking the accurateness of extracted data of random studies (N = 40; no mistakes detected) to minimize the chance for errors, although this does not fully guarantee 100% accuracy. For this reason, we also share the articles included of the different, the final set of articles, and the extracted statistics for the data analysis on OSF (https://osf.io/dgxsz/) to transparently share the decisions.

Results

AUC characteristics

In total, we extracted 1580 AUC values out of the 157 articles (see Table 1). Of these 1580 AUC values, 755 were full, 823 were partial, and 2 were difference AUC values. For 702 (44.1%) AUC values, the 95% CIs were reported but for the remaining 878 AUC values they were not. We also examined how the AUC values were interpreted. Results showed that 1,425 of these values (90%) were interpreted using statistical significance wherein it was either established that the AUC was greater than chance performance (AUC = .5) or was greater than another AUC. Of the remaining AUC values, twenty-five (1.6%) were interpreted using benchmarks, fifteen (.9%) were contextualized to the study at hand, two (0.1%) were interpreted subjectively (i.e., mention of magnitude without the use of benchmarks or contextualization), and 113 were not interpreted at all.

Table 1.

Characteristics of the AUC Values.

Type of AUC	***N* (%)**
Partial	823 (52%)
Full	755 (48%)
Difference	2 (0.1%)
95% CI	
Not reported	878 (56%)
Reported	700 (44%)
Report graph	2 (0.1%)
Interpretation of AUC	
No interpretation/statistical significance	1,425 (90%)
No interpretation	113 (7.2%)
Benchmarks	25 (1.6%)
Contextual	15 (0.9%)
Subjective	2 (0.1%)

Note. In total 1580 AUC values were extracted. Data are presented in descending order. Some articles reported both partial and full AUCs.

Full AUC

To examine the distribution of the full AUC values, we created a histogram in Figure 5 (for the interactive plot see https://osf.io/dgxsz/). The distribution is approximately normally distributed. Vertical lines are placed to indicate the benchmarks of .7, .8, .9 (Mandreker, 2010). The mean of the full AUC value is .71, 95% CI [.70, .72] with a median of .71, 95% CI

[.70, .72]. Visual examination of the histogram shows excesses and shortfalls above and below the .7 and .8 benchmarks but not for the .9 benchmark.

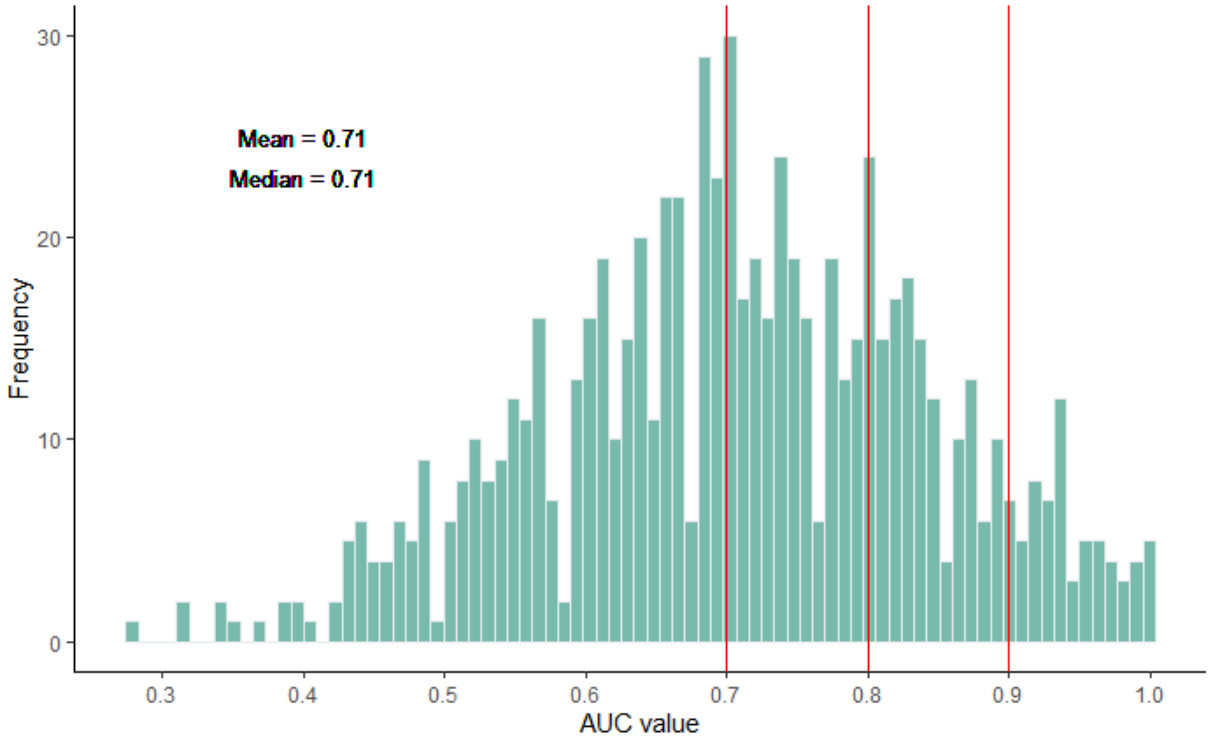


Figure 5. Histogram of full AUC values. Each red line corresponds to the given AUC benchmarks as reported in Mandreker (2010). Interactive histogram can be found on OSF (https://osf.io/dgxsz/).

Partial AUC

We also created a histogram in Figure 6 to visualize the distribution of the pAUC values (for the interactive plot see https://osf.io/dgxsz/). The mean of the pAUC values was .11, 95% CI [.10, .12] and the median was .08, 95% CI [.06, .09].

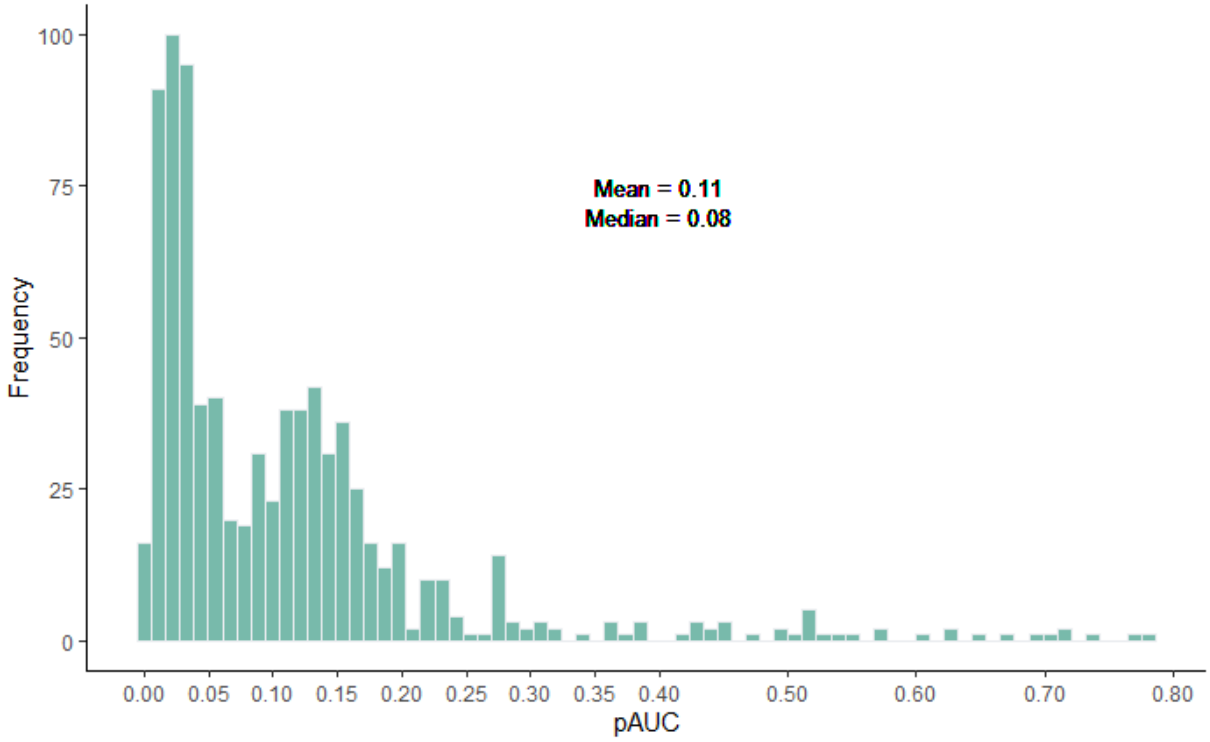


Figure 6. Histogram of partial AUC values. Interactive histogram can be found on OSF (https://osf.io/dgxsz/).

Power Analyses

We also examined how often a power analysis was conducted for each experiment and whether they were reproducible. In total, 239 experiments were conducted in the 157 articles. Out of the 239 experiments, 172 (72%) did not conduct a power analysis, 65 (27%) did conduct a power analysis, and 2 (.8%) conducted sensitivity analyses. Out of the 67 experiments that did provide a power analysis (or sensitivity), 19 were reproducible when certain assumptions were made (e.g., used 80% power, alpha = .05, or a one or two-tailed), and three were reproducible. The mean sample size was 1,937 (*SD* = 5,379), 95% CI [1,243, 2,631] with a median of 381, 95% CI [202, 610], and range of 20 - 39,471.

Table 2.

Characteristics of Power Analyses	
Power Analysis Conducted	
No	172 (72%)
Yes	65 (27%)
Sensitivity	2 (0.8%)
Power Analysis Reproducible	
No	45 (67%)
Yes, with assumptions	19 (28%)
Yes	3 (4.5%)
Sample Sizes	
Mean (SD)	1,937 (5,379)
Median (IQR)	381 (80, 1,584)
Range	20 - 39,471

Note. Data in rows associated with "Power Analysis Conducted" and "Power Analysis Reproducible" are presented in frequency (and percentage). SD stands for standard deviation. IQR stands for interquartile range.

Z-curve Analysis

We extracted the p-values for all ROC or AUC related analyses reported in the articles. In total, we extracted 819 p-values of which 474 were statistically significant and we conducted a Z-curve analysis (Bartoš & Schimmack, 2022). We observed an observed discovery rate of .58, 95% CI [.54, .61], an expected discovery rate of .62, 95% CI [.52, .73], and an expected replicability rate of .66, 95% CI [.57, .75] (see Figure 7).

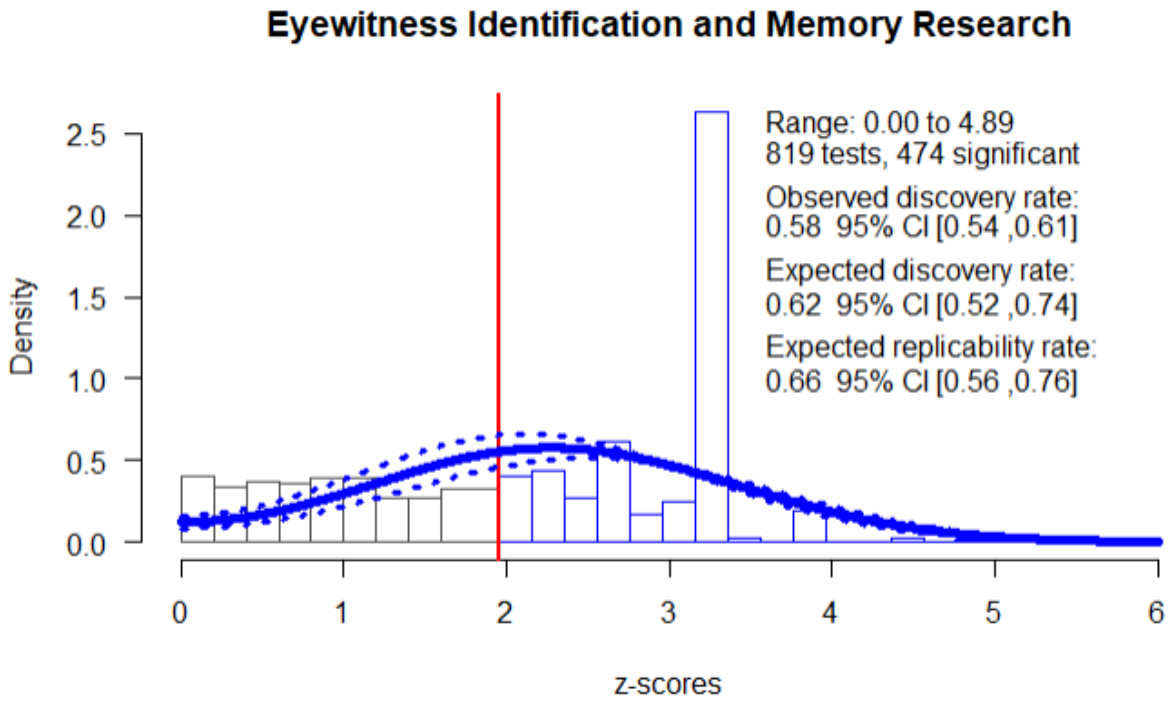


Figure 7. Z-curve analysis of the eyewitness identification and memory research.

Discussion

Because of the applied nature of eyewitness memory research, it is imperative to understand how practical recommendations are postulated in this type of research. An important feature of understanding the practical relevance of a study is by examining the magnitude of the phenomenon (i.e., effect size; Funder & Ozer, 2019; Otgaar et al., 2023). In eyewitness memory research, ROC curves and AUC values are frequently reported statistics designating the magnitude of a certain phenomenon. In this study, we examined how ROC curves and AUC values are used and interpreted in both the planning and data analysis stage of eyewitness memory research. More specifically, we were primarily interested how ROC curves and AUC values findings are translated into practical recommendations.

Similar to the examination of clinical prediction models (White et al., 2023), we found excesses of AUC values above and shortfalls below the proposed benchmarks (i.e., AUC = .7 - .8 is acceptable; .8 - .9 is good; Mandreker, 2010) for AUC values. This might suggest the

presence of questionable research practices to obtain such acceptable or good AUC values. However, almost no studies used these benchmarks to interpret their findings. Instead, the vast majority relied on statistical significance to interpret the meaningfulness or magnitude of their findings. This is in line with a recent survey wherein a substantial amount (13-40%) of memory researchers conflated statistical significance with practical relevance (Riesthuis et al., 2021). Relying exclusively on statistical significance can be problematic when sample sizes increase because this may lead to finding statistically significant differences for trivial findings (Anvari & Lakens, 2021; Kovera & Evelo, 2021). That is, with increasing sample sizes, statistically significant differences of 0.01 in (p-)AUC values can be detected which was the case in several of the included studies. This may be an important difference but explanations for why such a relatively small difference is relevant was frequently not reported (Anvari et al., 2023; for a medical example see Baker et al., 2014). Across all studies, the mean sample size for each study was 1,937[4] (median = 308) in which some studies had approximately twenty to forty thousand participants. With, for example, 20,000 participants (n_{cases} = 10,000, $n_{control}$ = 10,000) researchers would be able to find statistically significant AUC values of .51 (compared with AUC = .5) with statistical power (1 - β) of .95, and an α = .05 (see https://osf.io/dgxsz/ for power analyses). With only 2,000 participants, they would be able to find statistically significant AUC values of .55. These simplified power analyses are intended to show that with increasing sample sizes, statistically significant AUC values can be found that may be trivial.

The fact that most research relied on statistical significance to interpret their findings might explain why many did not use AUC benchmarks. Researchers might have been unaware about such benchmarks. Alternatively, researchers might have known that relying on benchmarks is problematic because determining whether an AUC is fair, good, excellent, or

[4] Data were skewed to the right. 95% CI of the mean is rather large and thus the median gives a better representation of the data.

practically relevant depends on the context and its associated costs and benefits (Baker et al., 2014; Lakens, 2022; Panzarella et al., 2021). For instance, if a lineup procedure has no costs (e.g., provide a minor instruction to focus on eyes before the lineup) but leads to an improved AUC (compared to the lineup without such instruction), even if relatively small, might already be good enough.[5] On the other hand, if the procedure is very costly or difficult (e.g., construct a 16 person versus 6 person fair lineup), greater differences in the ROC curves and AUC values might be warranted. However, only 15 AUC values from two studies were interpreted in light of the context of the study[6] which makes it unlikely that most researchers were aware of the importance of contextualizing their findings to understand the magnitude or meaningfulness of the results.

Smallest Effect Size of Interest in Eyewitness Memory Research

To gauge the practical meaning of a finding, it is important to establish what the smallest effect size of interest (SESOI) is for a specific research area (Lakens et al., 2018). The SESOI is the smallest effect size that is of interest to researchers, has practical implications, or is theoretically relevant. In eyewitness memory research, discussions on what the SESOI should be for practical implications are currently ongoing (for approaches on setting the SESOI see Lakens et al., 2018; Riesthuis et al., 2021; Otgaar et al., 2022; 2023). For instance, Riesthuis and colleagues (2021) recently asked memory scientists what they considered the SESOI to be for false memory research. To do so, memory scientists were

[5] Of course, tests between AUC values/ROC curves should be conducted to separate noise from signal.
[6] One example of a contextualized interpretation of one of the studies is: "Assuming a culprit-present base rate of 50%, the present findings suggest the rule out procedure (AUC = .8714) would lead to about 5 more correct classifications per 100 cases than does the traditional simultaneous lineup (AUC = .8264). When one considers the relevant context, that effect size is more impressive than it might seem at first blush. First, even an investigator who was flipping a coin would be able to correctly classify the suspect 50 times out of 100, so the maximum possible improvement is only 50 (and not 100). Second, because the rule out procedure follows the traditional lineup, it could not possibly have improved decision accuracy. It seems likely that the rule out procedure will only improve performance in instances in which the witness does not initially identify the suspect. Yet, simply by asking the witness for a few additional confidence ratings, police investigators might increase their ability to discriminate guilty suspects from innocent suspects.". (see https://osf.io/dgxsz/ for all interpretations).

presented with several hypothetical and real experiments examining, for example, the effects of a certain therapy on false memory formation. Then, they had to indicate how many more false details participants who received the therapy should misremember (versus participants who did not receive therapy) to imply that the therapy leads to unreliable testimonies. There was no general consensus among the memory scientists on what the SESOI should be and many relied on statistical significance or benchmarks to decide whether results were practically relevant.

Another approach that has been undertaken to establish the SESOI for eyewitness memory research was through a cost-benefit analysis wherein Otgaar and colleagues (2023) argued that the misremembering or forgetting of one detail could serve as the SESOI. The proposal of one detail was chosen because there are legal cases wherein one such memory error had severe legal consequences (e.g., see case of Ronald Cotton; Thompson-Cannino et al., 2009). In their discussion concerning possible SESOIs, they used unstandardized effect sizes (e.g., true and false identification rates; number of details remembered) because they are more robust, easier to calculate, but most importantly because they are simply more intuitive than standardized effect sizes (Baguley, 2009; Riesthuis et al., 2021).

A focus on unstandardized effect sizes can also facilitate the discussion on the SESOI for ROC curve analyses and AUC values in eyewitness identification research. That is, although AUC values provide a way to assess the discriminability of, for example, a lineup procedure between the true positive rate (i.e., correct identification rate) and false positive rate (i.e., incorrect identification rate), assessing whether this is sufficiently to base practical recommendations on depends on the context. In the case of eyewitness identification research, larger AUC values might be demanded as the costs of a false positive (e.g., innocent suspect going to jail) but also missing true positives (e.g., guilty person roaming free) are substantial. Based on true and false positive rates, the SESOI can be determined and justified via, for

example, using the Blackstone ratio which entails that "it is better that ten guilty individuals escape, than that one innocent suffer" (Blackstone, 1765). The underlying thought of the Blackstone ratio is also a cost-benefit analysis wherein the costs of an innocent individual being imprisoned are, for example, that the guilty perpetrator is roaming free, trust in justice system is reduced, and possible exoneration expenses. Blackstone argued that these costs outweigh the possibility of 10 guilty individuals escaping. This ratio could serve as guidance for determining the SESOI in terms of the smallest acceptable hit and false positive rates in the ROC curve analysis. However, research is necessary to decide on the smallest acceptable ratio and translating this into a SESOI for eyewitness identification research. For these reasons, we urge researchers in the field eyewitness identification research to start and continue the discussion on possible contextualized SESOIs as this will facilitate the interpretation of results and allow for equivalence (i.e., establishing there is no meaningful effect; Bradley, 2013) and minimum-effects testing (establishing practical significance; Murphy & Myors, 1999).

For both equivalence and minimum-effects testing but also to examine the uncertainty of an estimate, 95% CIs are crucial (Jané et al., 2024). That is, using the 95% CI, it can be established whether an observed finding is too small to care about (i.e., 95% CI is smaller than the SESOI), practically meaningful (95% CI is greater than the SESOI; Westlake, 1972), or to understand the uncertainty around the estimates (relatively large 95% CIs). In our study, less than half of the studies (= 44.1%) provided 95% CIs alongside the reported AUC values. However, when 95% CIs were reported, it was for the AUC value while 95% CIs along the ROC curves were never reported. However, it is possible to plot such 95% CIs for the entire ROC curve via "pROC" in R and we recommend researchers to do so (for an example see Figure 8 and R code on https://osf.io/dgxsz/). The visualization of the ROC curve and its 95% CIs is crucial to see whether ROC curves overlap. Specifically, if two ROC curves overlap

(see Figure 3), AUC values become less informative and alternative or local (at individual thresholds) tests need to be performed (for alternative analyses see for example Moise et al., 1988).

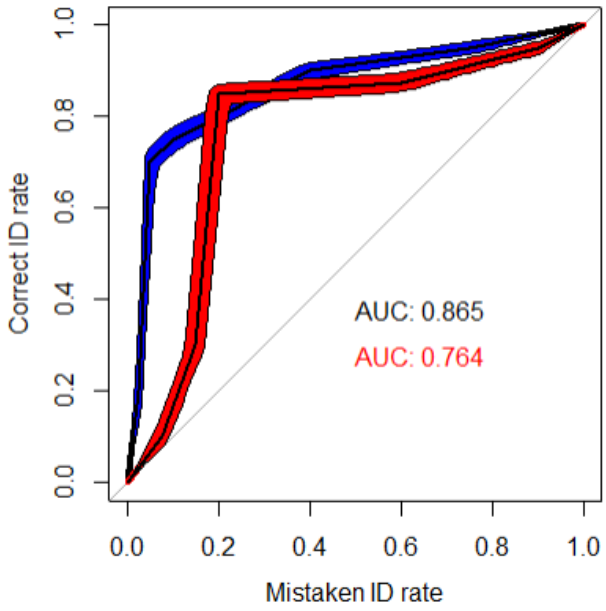


Figure 4. Example of ROC curves with 95% CI around the curves.

Establishing a SESOI will also improve the planning of new studies such as when conducting power analyses. That is, via power analyses researchers will be able to accurately estimate the number of participants necessary. This can inform researchers whether they have adequate funding to conduct a meaningful study or save funding by not collecting more data than necessary. Interestingly, similar to other fields (e.g., motor behavior research; McKay et al., 2022; Mesquida et al., 2023), the majority of eyewitness identification studies did not conduct power analyses. Moreover, when power analyses were conducted, they were oftentimes not reproducible. These findings highlight the need for researchers to conduct power analyses and make them reproducible just as their data and analyses (Lakens, 2022; Hardwicke et al., 2022). Although power analyses for ROC curve analyses and AUC values can be conducted via the R package "pROC" (Robin et al., 2011), further work is necessary to allow more contextualized power analyses using for example the SESOI (but see Mah, 2022).

Publications Bias and Limitations

We did not find any evidence of a publication bias in the eyewitness memory literature.[7] However, the expected replication rate was rather low (= .66, 95% CI [.57, .75]) which means that if exact direct replications of these studies were to be conducted, approximately 66% would be expected to find another statistically significant effect. It is important to note that some of these findings can be explained by the reporting of p-values. Frequently, when p-values were below .05 or .001, the exact p-values and corresponding test statistics (e.g., *t* or *F* values) were not reported. This issue was most apparent for p-values below .001 as a steep increase of z-scores around 3.2 can be observed in Figure 7. In this study, we only extracted the p-values but the outcomes of the z-curve analysis might differ when more exact values are included.

One limitation is that we only extracted whether studies reported 95% CIs for their AUC values, even though some studies provided other indices of variation (e.g., standard deviation, standard error). We only extracted the 95% CIs because they are typically the estimates of variation used to conduct equivalence tests and/or minimum-effects test (e.g., Lakens et al., 2018). That is, 95% CIs directly facilitate the interpretation of the AUC value in terms of practical meaningfulness. However, it is possible that researchers provided sufficient data to calculate the 95% CIs or that they interpreted the other indices of variation more meaningfully.

Conclusion

In this study, we examined how ROC curve analyses and AUC values are reported and interpreted in eyewitness memory research. The vast majority of studies relied on statistical significance to interpret the magnitude or meaningfulness of the findings. Relying on statistical significance might be especially problematic in this field of research as the sample

[7] Point estimate and 95% CI of observed discovery rate lies (= .58, 95% CI = .54, .61) within the 95% CI of the expected discovery rate (= .52, .73).

sizes are rather large which means that possible trivial findings are given more importance than is justified. Moreover, more than half of the studies did not report 95% CIs. Finally, many studies did not conduct power analyses and when they did, they were oftentimes not reproducible. We recommend discussions on establishing a SESOI, visually inspecting ROC curves with 95% CIs to examine whether AUC comparisons are warranted, and the need to conduct power analyses to improve the practical inferences of eyewitness memory research.

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